Hyperspectral Estimation of Community-Weighted Mean Traits in Wetland Plant Communities

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***Abstract*—Plant functional traits reflect plant resource-use strategies and responses to environmental stress and, thus, provide an important entry point for understanding wetland ecological processes. However, traditional field-based approaches to measuring functional traits are constrained in spatial coverage and efficiency by the poor accessibility and complex environmental conditions of marsh wetlands. In this study, three typical marsh wetlands within the Crested Ibis National Nature Reserve were selected as study sites. A UAV-borne hyperspectral sensor was used to acquire canopy spectral information, and quadrat surveys were conducted to determine wetland plant species composition. Six functional traits, including leaf area, leaf thickness, plant dry matter content, and leaf carbon, nitrogen, and phosphorus contents, were measured and converted into community-weighted mean traits (CWMs). Using vegetation indices and fractional-order differential spectral features, we developed hyperspectral estimation models for the CWMs. The results showed that: 1) fractional differentiation of order 0.50–1.25 substantially enhanced spectral feature extraction; 2) hyperspectral data showed good potential for estimating CWMs in wetlands, with optimal coefficient of determination (R^2) values ranging from 0.56 to 0.64; and 3) different CWMs exhibited trait-specific responses to spectral features and models. This study provides a technical foundation for efficient, fine-scale monitoring of wetland plant functional structure and supports ecological conservation and management of wetland ecosystems.**

***Index Terms*—Community-weighted mean traits (CWMs), fractional-order differentiation, hyperspectral, machine learning.**

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I. Introduction

PLANT functional traits are defined as functional characteristics that have a significant influence on plant establishment, growth, and survival strategies. Acting individually or in combination, these traits reflect plant adaptive strategies to environmental change and serve as important indicators of ecosystem functioning [1]. Trait-based approaches provide deeper insight into species interactions, coexistence mechanisms, and the constraints on species' persistence under environmental stress and are, therefore, of great importance for elucidating the mechanisms underlying wetland ecosystem functioning and for maintaining wetland health and stability [2].

Conventional measurements of plant functional traits rely primarily on field surveys. Although such methods can produce highly accurate trait values, they are time-consuming and labor-intensive, and the complexity and limited accessibility of wetland environments have become major constraints on trait investigations over large spatial extents. Previous studies have shown that plant functional traits influence the processes of light absorption and scattering, and differences in traits lead to differences in plant optical properties, thereby causing variation in measured spectra [3], [4]. For example, trichome density and epidermal thickness have been identified as the most important factors affecting the reflectance spectra of wetland species [5], while, in aquatic plants, chlorophyll a, chlorophyll b, and specific leaf area (SLA) together can explain about 60% of the variability in reflectance spectra. Further research has shown that biochemical traits play an even more important role in explaining spectral variation in aquatic plants [6]. Therefore, the strong explanatory power of wetland plant functional traits for spectral variation provides a basis for monitoring these traits using remote sensing. Hyperspectral remote sensing, in particular, provides spectrally detailed and information-rich data and enables rapid, nondestructive, repeatable, and large-area observations, making the fast and accurate retrieval of wetland plant functional traits feasible [4], [7], [8].

From a methodological perspective, the estimation of wetland plant functional traits has mainly relied on statistical approaches and machine learning models. Among statistical methods, partial least-squares regression (PLSR) is one of the most representative techniques [9], [10]. For example, previous studies have developed a biomass estimation model for *Suaeda salsa* based on hyperspectral data using PLSR,

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achieving a coefficient of determination (R^2) of 0.85 [11], and have further demonstrated, via polynomial regression, the potential of hyperspectral techniques for estimating chlorophyll fluorescence parameters of *Suaeda salsa* under salinity and water stress [12]. In addition, an established method involves proposing eight narrowband hyperspectral vegetation indices sensitive to SPAD values, selecting sensitive wavelengths via the successive projections algorithm (SPA), and establishing simple linear regression models between these indices and SPAD values [13]. Machine learning regression models can generally be divided into two categories: single models, such as support vector machine regression (SVM), and ensemble models, such as random forest (RF) and extreme gradient boosting (XGBoost). By combining RF with continuous wavelet transform and integrating species information and soil physicochemical properties, the estimation accuracy of leaf SPAD values, water content, nitrogen concentration, and phosphorus concentration in mangrove species can be significantly improved [14]. Moreover, studies that have applied multiple machine learning regression models to the estimation of functional traits of *Spartina alterniflora* have shown that XGBoost, when using first-derivative spectra as input, yields the highest prediction accuracy [15].

From a scaling perspective, the estimation of wetland plant functional traits has been explored at both the leaf and canopy levels. Leaf-scale remote sensing of plant functional traits started relatively early and has mainly relied on handheld portable spectrometers [16]. For example, correlation analysis and stepwise regression have been used to demonstrate that hyperspectral techniques can effectively estimate leaf chlorophyll and total nitrogen contents in wetland plants [17], and leaf morphological traits and ecological stoichiometric characteristics can also be accurately estimated using hyperspectral data [18], [19]. At the canopy scale, remote sensing data can be acquired through a variety of platforms, including handheld field spectrometers, imaging spectrometers mounted on airborne platforms, and imaging spectrometers mounted on satellite platforms [20]. Field spectroradiometers provide richer hyperspectral information and, when combined with derivative transformations and machine learning models, can effectively estimate functional traits such as total carbon, total nitrogen, and biomass [21]. In addition, uncrewed aerial vehicles (UAVs) have offered new opportunities for canopy-scale studies and have been shown to have significant advantages in vegetation monitoring [22], [23]. Using UAV-based hyperspectral imagery, biomass estimation models for aboveground biomass (AGB) stratified by vegetation type have been developed, and the results indicate that RF models can effectively estimate the biomass of dominant wetland species [24].

In this study, we first calculated 16 commonly used classical vegetation indices and applied fractional-order differentiation to transform the spectral features of the hyperspectral imagery. Spearman's rank correlation analysis was then used to extract spectral variables that were significantly correlated with the community-weighted mean traits (CWMs). Two modeling strategies were designed: 1) CWMs' trait estimation models based solely on vegetation indices and 2) CWMs' trait estimation models based on fractional-order differential spectra. Four regression algorithms, PLSR, genetic algorithm (GA)-optimized PLSR, RF, and XGBoost, were employed to evaluate the prediction accuracy of CWMs, and the best-performing models were subsequently used for the estimation of CWMs of wetland plant functional traits.

II. Materials and Methods

A. Study Area

The Crested Ibis National Nature Reserve (107°21′E–107°44′E, 33°08′N–33°35′N) is located in Hanzhong City, Shaanxi Province, China, and serves as the core habitat of the wild Crested Ibis (*Nipponia nippon*). The reserve covers an area of 375.49 km^2 and extends across Yang County and Chenggu County (see Fig. 1). It is situated in the transitional zone between the southern foothills of the Qinling Mountains and the northern margin of the Bashan Mountains and is characterized by an East Asian monsoon climate transitional from northern subtropical to warm temperate, with distinct seasons and a marked coincidence of heat and rainfall.

The river network in the reserve belongs to the Han River system and exhibits relatively high drainage density, providing favorable hydrological conditions for the formation and maintenance of wetlands. The dominant soil types include fluvo–aquic soil, yellow–brown soil, brown soil, paddy soil, and alluvial soil. Vegetation is highly diverse, with approximately 1500 vascular plant species belonging to 135 families and 600 genera. Wetlands within the reserve are mainly composed of marsh wetlands and floodplain wetlands. In this study, three typical marsh wetlands (Plot1, Plot2, and Plot3) were selected as sampling sites. These marshes are dominated by perennial hygrophilous herbaceous species, primarily *Oenanthe javanica*, *Alternanthera philoxeroides*, *Echinochloa crus-galli*, *Sagittaria trifolia*, and *Equisetum hyemale*.

B. Data Acquisition

First, canopy-scale spectral information over the study area was acquired. UAV flights were conducted under clear, cloud-free, and low-wind conditions, with flight times constrained to 11:00–14:00 to minimize the influence of solar zenith angle on data quality. Hyperspectral data were collected using a Tianyuan Ourui hexacopter equipped with an S185 high-speed imaging spectrometer (Cubert, Germany), ensuring full coverage of the three plots. The spectral range of the image is 450–950 nm, comprising 135 bands, with a spectral resolution of 4 nm (see Table S1).

Following the airborne survey, a total of 150 quadrats were established across the three plots according to a typical sampling design, with 45, 49, and 56 quadrats in Plot1, Plot2, and Plot3, respectively. Each quadrat measured 1 m^2, and the horizontal distance between neighboring quadrats was maintained at no less than 10 m. For each quadrat, all plant species and their individual counts were recorded in detail. For each species within a quadrat, three intact leaves without visible mechanical damage were collected from the upper, middle, and lower parts of the plant (one leaf per position), and three

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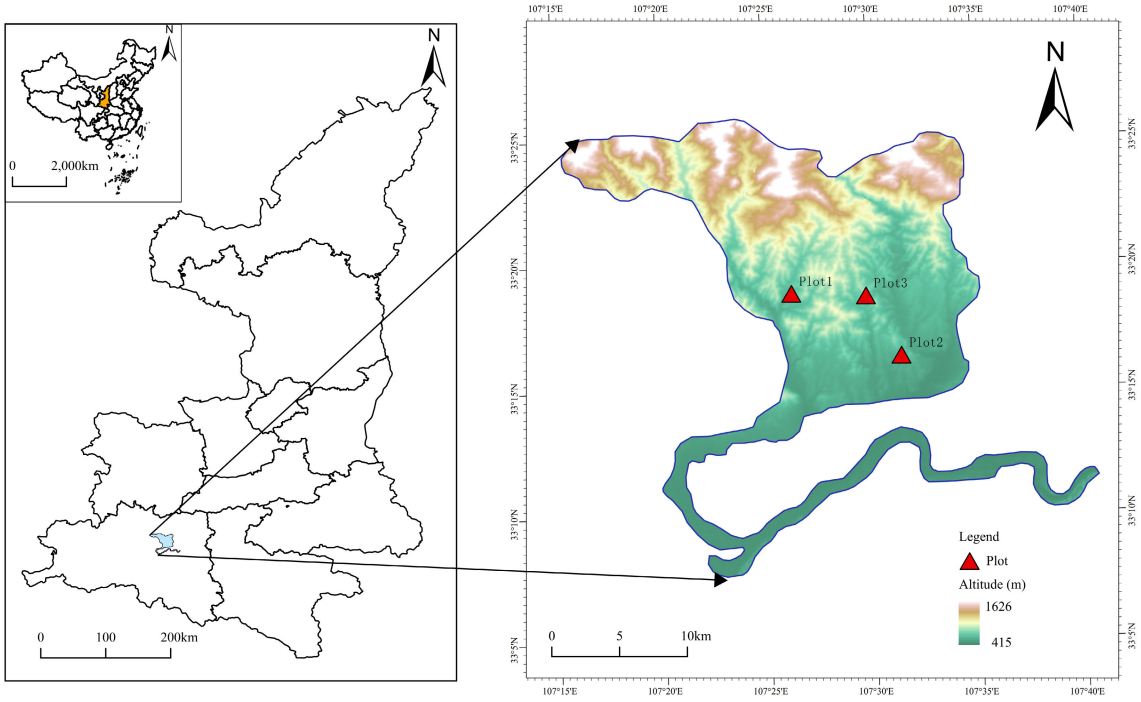


Fig. 1. Study area distribution.

TABLE I
STATISTICAL RESULTS OF CWMs

Community-weighted mean traits	Minimum	Maximum	Mean	Standard deviation
CMW_LA (cm²)	1.33	27.59	11.06	4.30
CWM_LT (cm)	0.03	0.86	0.42	0.19
CWM_PDMC (g/g)	0.22	0.44	0.33	0.04
CWM_TC (g/kg)	351.84	442.27	394.81	14.06
CWM_TN (g/kg)	12.45	25.24	18.53	2.68
CWM_TP (g/kg)	2.29	5.61	3.82	0.69

whole individuals per species were clipped at ground level. All leaf and whole-plant samples were immediately sealed in plastic bags and stored in a portable incubator for subsequent measurements of leaf area, leaf thickness, plant dry matter content, total carbon, total nitrogen, and total phosphorus. The geographic coordinates of each quadrat were recorded in the field using a handheld GPS device to enable accurate spatial matching between spectral data and field survey data.

Based on the measured plant functional traits, CWMs were further calculated. The CWMs were computed according to the following:

$$CWMs = \sum_{i=1}^{S} a_i \times t_i \tag{1}$$

where a_i is the relative abundance of species i and t_i is the functional trait value of species i. In this study, we calculated community-weighted means for leaf area (CWM_LA), leaf thickness (CWM_LT), plant dry matter content (CWM_PDMC), total carbon (CWM_TC), total nitrogen (CWM_TN), and total phosphorus (CWM_TP).

The statistical results of all samples are shown in Table I. Then, for each trait, all samples were sorted in descending order, and the sequential selection method was used to divide the samples into training and validation sets at a ratio of 2:1.

C. Preprocessing and Feature Extraction of Hyperspectral Data

1) Environmental Background Filtering: To minimize the interference of environmental background effects, an NDVI threshold-based filtering procedure was applied before spectral analysis. Specifically, pixels with NDVI values below 0.4 were removed, as these pixels generally represented nonvegetated surfaces, sparse vegetation, or mixed background information, such as soil, water, and shadows [25], [26]. Only pixels with NDVI ≥ 0.4 were retained for subsequent spectral feature extraction and model construction, thereby reducing background noise and enhancing the vegetation-related spectral signal.

2) SG Filtering: The Savitzky–Golay (SG) filtering method is primarily used for noise reduction and smoothing of hyperspectral remote sensing data. Compared with traditional

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moving-average filters, it has a pronounced advantage in preserving the characteristic features of the original signal. SG filtering fits a local polynomial to the data within a moving window using the least-squares method to derive a set of coefficients c_k for each data point. After smoothing, the value at the jth wavelength can be calculated using as follows:

$$x_{j,SG} = \frac{1}{N}\sum_{k=-m}^{+m} c_k x_{j+k} \tag{2}$$

$$N = \sum_{k=-m}^{+m} c_k \tag{3}$$

where m is the number of data points on one side of the window at a given wavelength and N is the normalization factor. In our study, the width of the smoothing window and the degree of the smoothing polynomial are set to 7 and 2, respectively [27].

3) Vegetation Index Calculation: In this study, 16 traditional vegetation indices were selected (see Table II) and calculated from the original hyperspectral data. These indices were subsequently used as predictor variables for CWMs' estimation.

4) Fractional-Order Differential Transformation: Fractional-order derivatives are an extension of classical integer-order differentiation and can sensitively capture subtle variations in spectral reflectance. They have been applied in remote sensing data processing and shown to be suitable for enhancing spectral features [28], [29]. At present, there is no single unified definition of fractional-order derivatives, and several formulations have been proposed, including those of Riemann–Liouville, Caputo, and Grunwald–Letnikov (G–L). Among these, the G–L form is the most widely used [30] and was also adopted in this study. We applied G–L fractional differentiation to hyperspectral reflectance data from order 0–2 with a step size of 0.25, implemented in MATLAB R2016a, and the G–L fractional derivative is expressed as shown as follows:

$$\frac{d^u f(x)}{dx^u} \approx f(x) + (-u) f(x-1) + \frac{(-u)(-u+1)}{2} f(x-2) + \cdots + \frac{\Gamma(-u+1)}{m!\Gamma(-u+m+1)} f(x-m) \tag{4}$$

where Γ denotes the Gamma function, x is the wavelength, m is the difference between the upper and lower limits of differentiation, and u is the order of the derivative.

5) Spearman Correlation: Spearman correlation analysis calculates the correlation coefficient after transforming the original data into ranks and is used to evaluate the monotonic association between two variables. In this study, Spearman's rank correlation coefficients ρ between spectral reflectance under nine fractional-order differential transformations and CWMs were computed using Python 3.11. For each CWM, only wavelength bands with $p < 0.05$ were retained as sensitive bands. The Spearman correlation coefficient ρ is given as follows:

$$\rho = 1 - \frac{6\sum d_i^2}{n\left(n^2-1\right)} \tag{5}$$

where d_i is the difference between the ranks of the two variables for the ith sample and n is the number of samples.

6) Recursive Feature Elimination: Recursive feature elimination (RFE) is a backward feature selection method. This method first trains a model using the full set of features and ranks the features according to model-derived feature importance. Less important features are then recursively removed, the model is retrained, and the optimal feature subset is finally determined based on model prediction performance [31]. In this study, based on the results of correlation analysis, RFE was combined with the RF model to gradually select the optimal feature subset according to the contribution of each feature to model prediction performance. During this process, tenfold cross-validation was used to evaluate the regression prediction performance of different feature subsets, and the root mean square error (RMSE) of the validation set was calculated. Finally, the feature subset with the lowest average RMSE among different feature subsets was selected as the optimal feature combination.

D. Construction and Accuracy Assessment of Estimation Models

Based on the selected key predictor variables, this study employed four modeling approaches, namely, XGBoost, RF, partial least-squares (PLS), and a GA-optimized PLS model (GA-PLS), to construct estimation models for wetland plant CWMs and functional diversity indices. To improve model performance and reduce potential overfitting, a random search strategy was used for local hyperparameter optimization. The specific hyperparameters and their search ranges are provided in Table S2.

The accuracy of the models was evaluated using three metrics: R^2, index of agreement (IOA), and the RMSE. R^2 quantifies the goodness of fit of the regression model and reflects the explanatory power of the independent variables for the dependent variable, with values ranging from 0 to 1. Values of R^2 approaching 1 indicate that the model effectively captures the relationship between predictors and response, whereas values close to 0 suggest poor model performance. RMSE measures the square root of the mean squared difference between observed and predicted values; smaller RMSE values indicate that model predictions are closer to the observations. IOA measures the degree of agreement between observed and predicted values, with values ranging from 0 to 1. A higher IOA indicates better agreement and stronger predictive performance. The formulas for calculating R^2, RMSE, and IOA are given as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - x_i)^2}{\sum_{i=1}^{n}(y_i - \bar{x}_i)^2} \tag{6}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - x_i)^2} \tag{7}$$

$$IOA = 1 - \frac{\sum_{i=1}^{n}(y_i - x_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y} + x_i - \bar{x}_i)^2} \tag{8}$$

where yi is the observed value of the ith sample, xi is the predicted value of the ith sample, $\bar{x}_i$ is the mean of the predicted values, and n is the number of samples.

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TABLE II
CALCULATION FORMULAS FOR 16 CLASSIC VEGETATION INDICES

Vegetation indices	Equation	Reference
Visible Atmospherically Resistant Index (VARI)	$(R_{550} - R_{660})/(R_{550} + R_{660} - R_{470})$	[40]
Optimized Soil-Adjusted Vegetation Index (OSAVI)	$1.16 \times (R_{880} - R_{670})/(R_{800} + R_{670} + 0.16)$	[40]
Modified Chlorophyll Absorption in Reflectance Index (MCARI)	$\big((R_{700} - R_{670}) - 0.2 \times (R_{700} - R_{550})\big) \times (R_{700}/R_{670})$	[40]
Triangular Vegetation Index (TVI)	$0.5 \times \big(520 \times (R_{750} - R_{550}) - 200 \times (R_{670} - R_{550})\big)$	[40]
Enhanced Vegetation Index (EVI)	$2.5 \times [(R_{800} - R_{270})/(R_{800} + 6 \times R_{270} - 7.5 \times R_{475} + 1)]$	[67]
Soil-Adjusted Vegetation Index (SAVI)	$(1 + 0.5) \times (R_{800} - R_{670})/(R_{800} + R_{670} + 0.5)$	[67]
Normalized Difference Vegetation Index (NDVI)	$(R_{850} - R_{675})/(R_{850} + R_{675})$	[68]
Normalized Difference Red Edge Index (NDRE)	$(R_{790} - R_{720})/(R_{790} + R_{720})$	[68]
Green Normalized Difference Vegetation Index (GNDVI)	$(R_{750} - R_{550})/(R_{750} + R_{550})$	[68]
Ratio Vegetation Index (RVI)	R_{790}/R_{670}	[69]
Green Chlorophyll Index (CIgreen)	$R_{780}/R_{550} - 1$	[70]
Transformed Chlorophyll Absorption in Reflectance Index (TCARI)	$3 \times (R_{700} - R_{670}) - 0.2 \times (R_{700} - R_{550}) \times (R_{700}/R_{670})$	[70]
Normalized Difference Index (NDI)	$(R_{850} - R_{710})/(R_{850} + R_{680})$	[70]
Red-Edge Chlorophyll Index (CIred)	$R_{800}/R_{740} - 1$	[70]
Ratio Spectral Index (RSI)	R_{825}/R_{735}	[70]
Water Adjusted Vegetation Index (WAVI)	$(1+\mathrm{L}) \times (R_{760-900} - R_{450-520})/(R_{760-900} + R_{450-520} + \mathrm{L})$	[59]

E. Shap Interpretability

Shapley values were originally proposed by Lloyd Shapley at the University of California to address the problem of profit allocation in cooperative games by fairly distributing payoffs among participants [32]. Specifically, Shapley values are obtained by considering all possible coalitions of players, computing the marginal contribution of each player to the total payoff in every coalition and then averaging these marginal contributions to derive the Shapley value of each participant.

In machine learning, Shapley values have been adapted as a model interpretation technique known as SHapley Additive exPlanations (SHAP). SHAP is used to quantify the contribution of each feature to the model prediction, that is, the average marginal contribution of each feature to the output, and is particularly suitable for complex nonlinear models.

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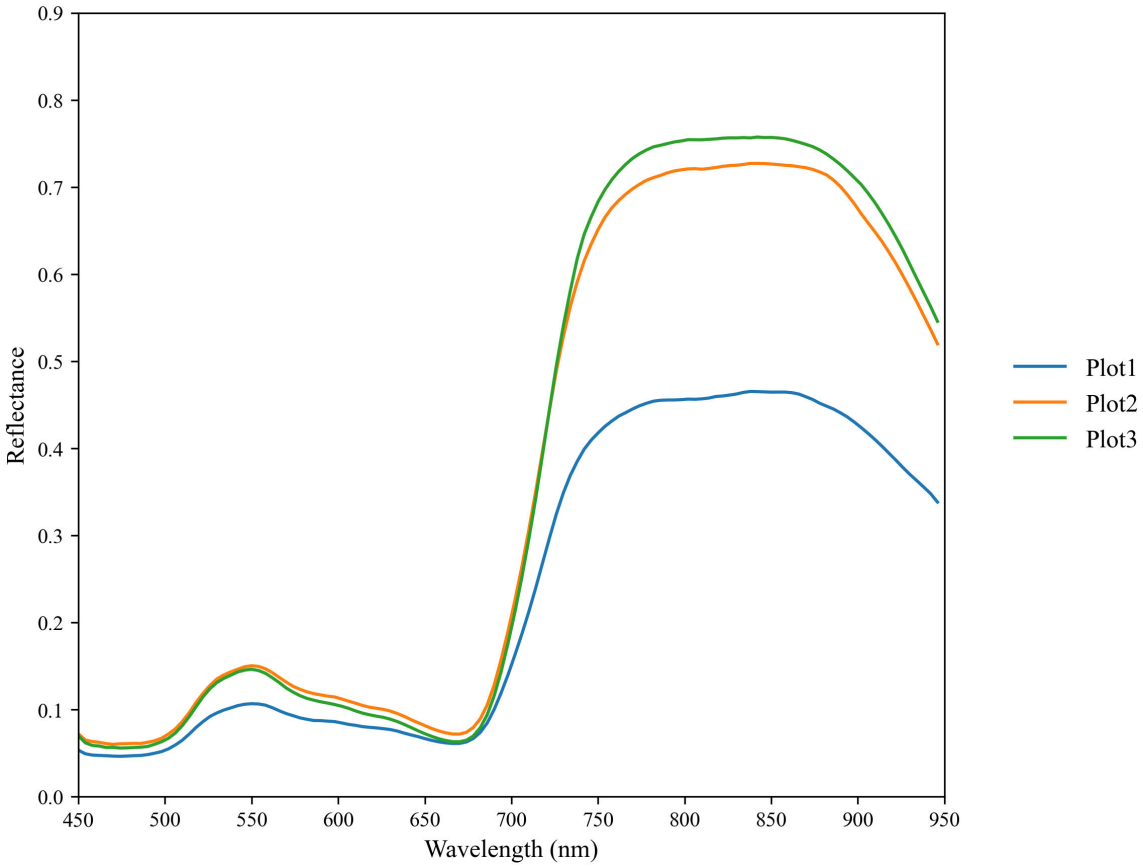


Fig. 2. Spectral curves of wetland plant communities.

An important advantage of this method is that it not only identifies which variables are important but also clarifies how these variables influence the final prediction [33].

In this study, SHAP analysis was implemented in Python 3.11 using the shap package, which provides a variety of visualization tools for displaying feature importance and illustrating the contribution of individual features to the predicted value of each sample.

III. Results

A. Hyperspectral Characteristics of Wetland Plant Community Canopies

Fig. 2 shows the raw spectral reflectance of plant community canopies in Plot1, Plot2, and Plot3 over the 450–950-nm wavelength range. Overall, the three plots exhibit similar spectral patterns, and all display the typical spectral characteristics of green vegetation canopies. In the green region, a small reflectance peak is observed around 550 nm, with reflectance magnitude following the order Plot3 $\approx$ Plot1 $>$ Plot2. Near 675 nm (in the red region), a slight absorption trough can be identified. Subsequently, in the near-infrared region (700–950 nm), all three plots show a pronounced reflectance peak, and the reflectance again follows the order Plot3 $\approx$ Plot1 $>$ Plot2. Overall, the canopy spectral reflectance of wetland plant communities in Plot2 is clearly lower than that in Plot1 and Plot3, which may reflect differences in vegetation types or canopy cover among the plots.

B. Spectral Characteristics and Feature Extraction Under Fractional-Order Differentiation

1) Canopy Spectral Characteristics Under Fractional-Order Differentiation: Fig. 3 illustrates the variation patterns of the original spectra under different fractional-order differential transformations. In the range from 0.25 to 0.75 order, the spectral curves become progressively more refined as the derivative order increases, and more subtle spectral variations emerge, particularly around 650 and 750 nm, where fluctuations become more pronounced. This indicates that low-order fractional differentiation can enhance key spectral features embedded in the data.

In the visible region, fractional-order differential transformations at 0.25–0.75 orders made the green reflectance peak near 550 nm and the chlorophyll absorption feature near 675 nm more distinct, thereby enhancing subtle spectral variations associated with vegetation pigments and biochemical composition. These enhanced features provided more effective spectral information for characterizing biochemical traits such as CWM_TC and CWM_TN. In the red-edge region, spectral variations within 680–720 nm are sensitive to vegetation physiological status and canopy structure. After fractional-order differential transformations at 0.50–0.75 orders, the local gradient variations in the red-edge region were further strengthened, improving the responsiveness of spectral features to morphological traits, such as CWM_LT and CWM_LA. When the order reached 1.00, the red-edge peak near 700 nm

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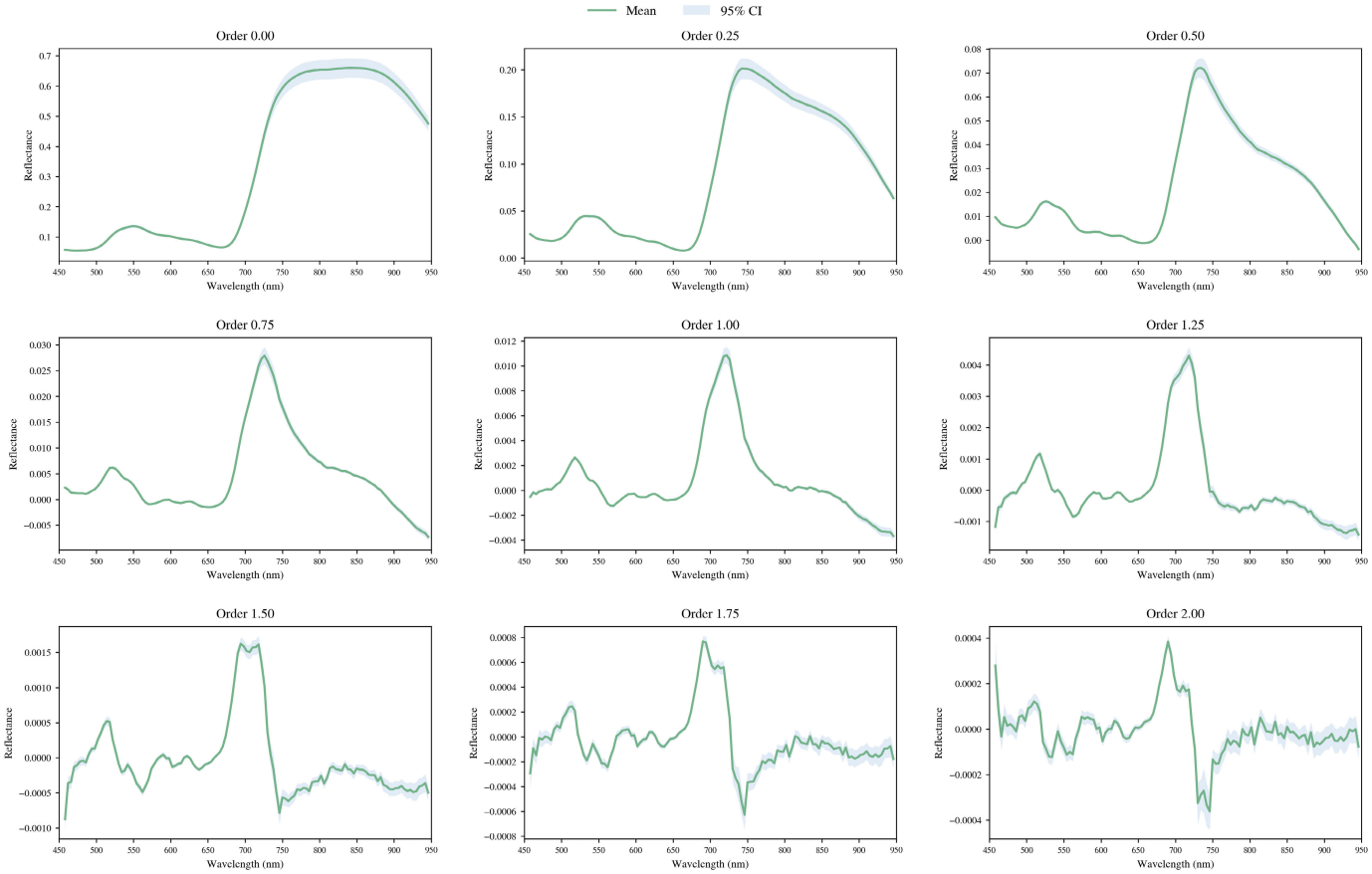


Fig. 3. Canopy spectral characteristics under fractional-order differential transformations.

became more prominent, indicating that this order could enhance local spectral variation information in the red-edge region to some extent. In the near-infrared region, spectra within 750–950 nm are mainly influenced by leaf internal structure, canopy structure, and water absorption. Low-order fractional-order differential transformations enhanced the local fluctuation features in this region, which may be associated with the estimation of CWM_PDMC.

2) Spectral Feature Extraction Under Fractional-Order Differentiation: Fig. 4 presents heatmaps of the correlations between spectra under different fractional-order derivatives and the CWMs. Overall, the composite traits exhibit similar correlation patterns across the various fractional orders. In general, all traits show strong positive correlations with spectral reflectance in the longer wavelength region (above 700 nm), and this positive correlation is further enhanced as the fractional order increases. In the middle wavelength region (500–700 nm), correlations are predominantly negative, particularly at lower derivative orders; as the order increases, some of these negative correlations weaken. In the shorter wavelength region, positive correlations are relatively consistent across most traits, indicating that spectral information in this region also contributes to trait prediction. These results suggest that by adjusting the fractional order, it is possible to enhance or attenuate the correlation between spectra and traits in different wavelength regions, which is particularly advantageous for capturing the relationships between canopy spectra and CWMs in the longer wavelength region.

Further analysis showed that for CWM_LA, the maximum correlation (0.61) occurred at 902–910 nm at orders 0.25–0.50; at orders 0.75–1.25, the peak shifted to 722–738 nm (0.55-0.56), indicating that sensitive spectral information concentrates in the red-edge to NIR region, and low-to-medium order differentiation enhances local reflectance features. For CWM_LT, the band of maximum correlation shifted with order: -0.55 at 646 nm (order 0.50) to 0.50 at 722 nm (orders 1.00–1.25), suggesting sensitivity to red-edge and middle-band gradients, with the best enhancement at medium orders. CWM_PDMC showed maximum correlation at 718–746 nm (absolute value 0.60) for orders 0.50–1.25, with the strongest response in the red-edge and NIR. CWM_TC peaked at 706–726 nm, with coefficients 0.72–0.73 across orders 0.25–1.25, indicating primary spectral information in the red-edge. CWM_TN exhibited a peak at 718–742 nm and coefficients 0.60–0.66 (orders 0.25–1.25) while maintaining negative correlations in 470–550 nm under low orders, reflecting indirect characterization via chlorophyll absorption; order 0.75 amplified subtle variations in the red-edge and visible bands. For CWM_TP, the maximum correlation band was 706–738 nm (0.60-0.61) at orders 0.50–1.25, and low-to-medium order differentiation enhanced both visible and red-edge sensitivity.

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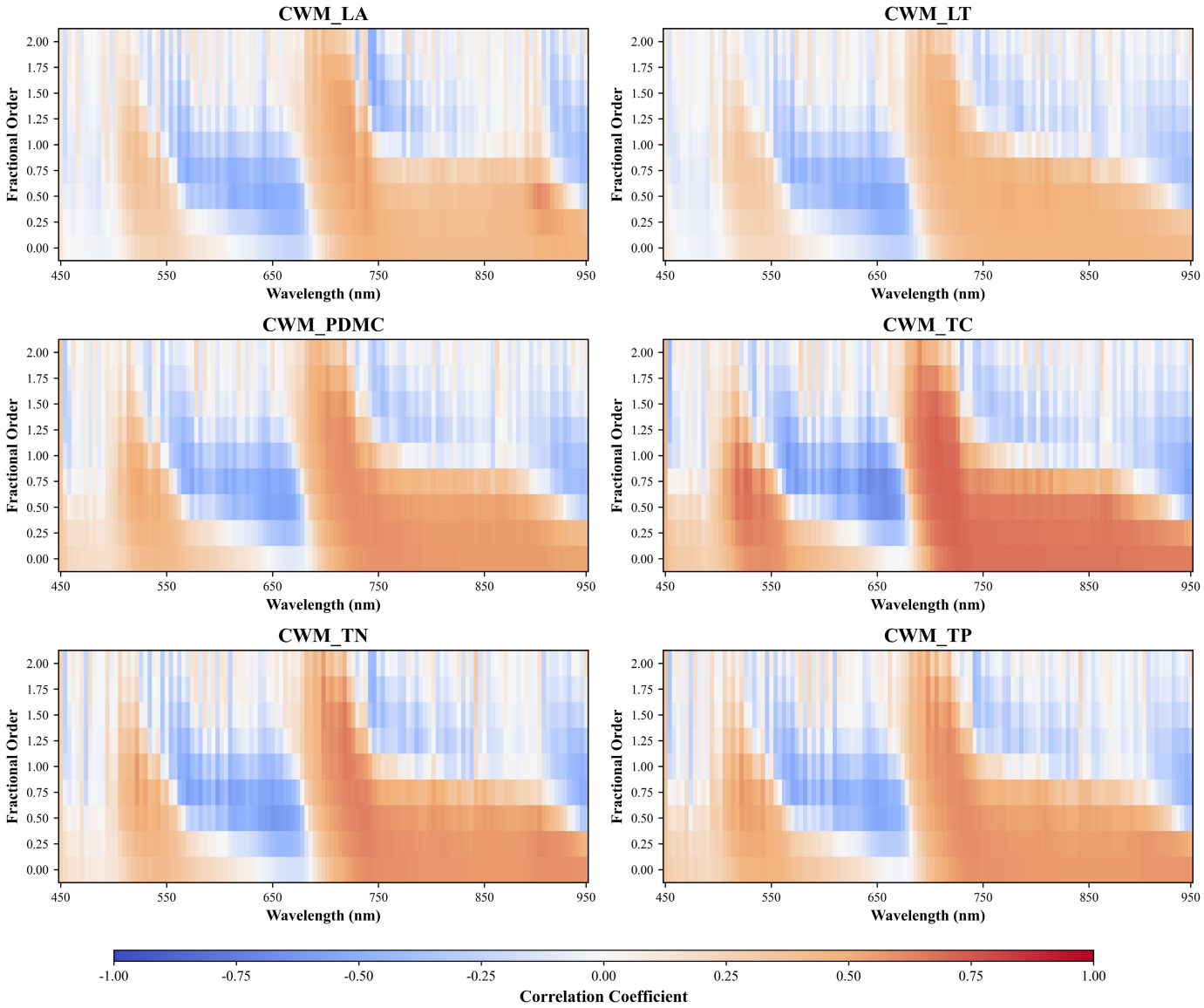


Fig. 4. Correlation analysis between spectra under different fractional-order differential transformations and CWMs.

Fig. 5 illustrates the dynamic variation in model prediction error, quantified by RMSE, as a function of the number of selected features during RFE following the initial correlation-based screening. Overall, the prediction errors for all CWMs exhibited a consistent pattern. In the initial stage with a small number of features (0–20), the RMSE decreased rapidly with increasing feature count, indicating a marked improvement in predictive performance. Beyond the range of 20–40 features, the rate of decline in RMSE slowed considerably, and the curves tended to stabilize, suggesting that the marginal contribution of additional features to prediction accuracy became limited, and model performance had reached a saturation state. Furthermore, for all CWMs, models employing low-order parameters, such as orders 0.00, 0.25, and 0.50, consistently achieved lower RMSE values at the convergence stage, thus delivering superior predictive accuracy. In contrast, models based on high-order parameters, such as orders 1.75 and 2.00, exhibited higher initial errors and maintained elevated error levels even after convergence. This indicates that low-order parameter settings are more conducive to enhancing model prediction performance.

C. Accuracy Evaluation of CWMs' Estimation Models

1) Accuracy Evaluation of Models Based on Vegetation Indices: Fig. 6 shows the estimation accuracy of the six CWMs under the four models, namely, PLS, GA-PLS, RF, and XGBoost. For CWM_LA and CWM_LT, the RF model yields relatively high R^2 and IOA values. For CWM_PDMC, the GA-PLS model performs the best, with R^2 of 0.55, while XGBoost shows the poorest performance, characterized by the lowest R^2 and IOA with the largest RMSE.

For biochemical traits, GA-PLS achieves R^2 values of 0.63 and IOA of 0.87 for CWM_TC, indicating relatively good predictive performance. For CWM_TN, the four models exhibit similar accuracy, with R^2 values around 0.52; among

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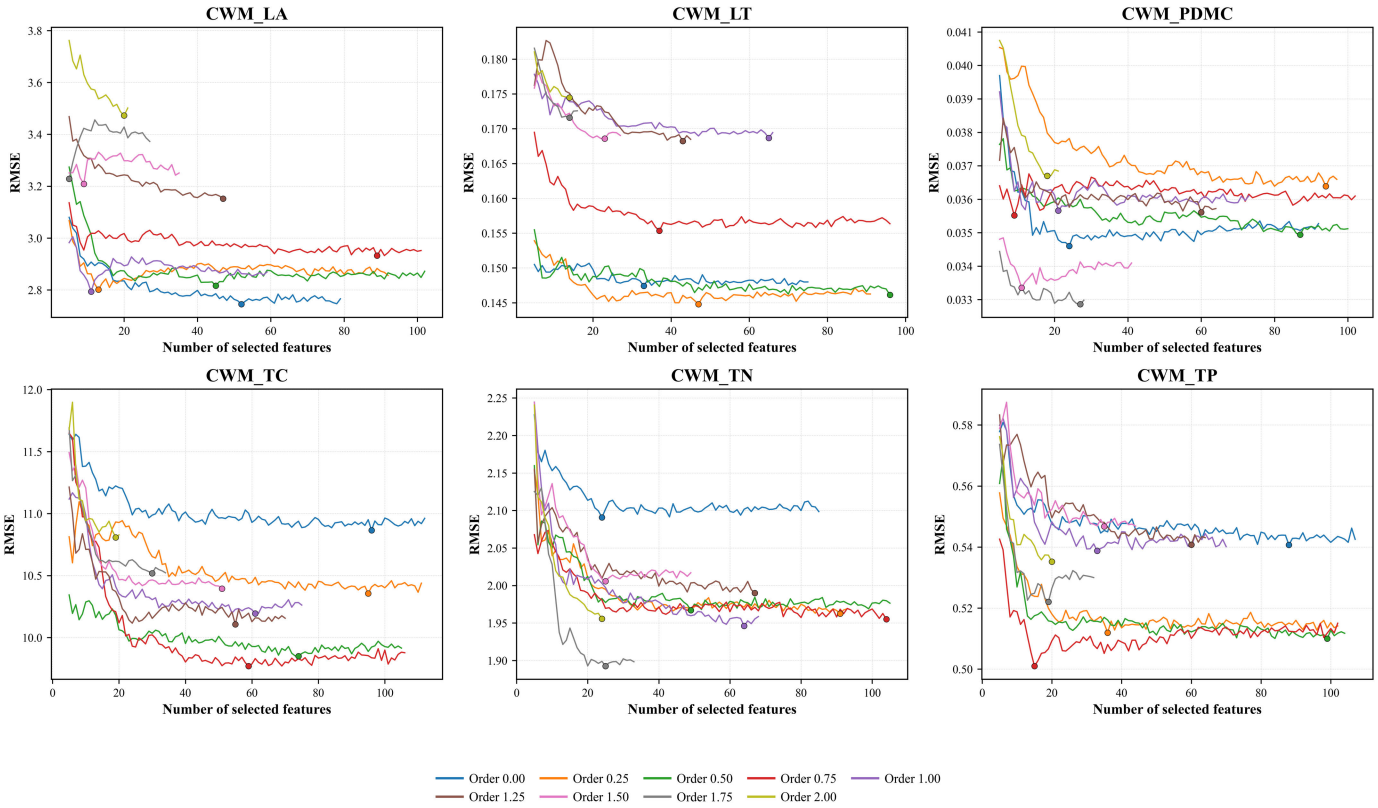


Fig. 5. Feature band selection results based on the RFE method under different fractional-order differential transformations.

them, RF performs slightly better, with R^2 = 0.54, IOA = 0.83, and RMSE values of 1.81. For CWM_TP, the RF model also achieves the highest accuracy (R^2 = 0.60, IOA = 0.85, and RMSE = 0.43), substantially outperforming the other three models.

Overall, the RF model provides the best predictions for most CWMs, particularly for CWM_LA, CWM_LT, CWM_TN, and CWM_TP, where it combines higher R^2 and IOA with lower RMSE. The GA-PLS model shows comparatively superior performance for CWM_PDMC and CWM_TC.

2) Accuracy Evaluation of Models Based on Differential Spectral Features: Figs. S1 and 7 show the estimation accuracy of the four models for the CWMs using feature bands extracted from correlation analysis and RFE, respectively, under different fractional-order derivatives. Across both feature selection strategies, low fractional orders consistently yielded the highest accuracy, as reflected by higher R^2 and IOA values and lower RMSE, while performance tended to decline with increasing order, particularly for orders above 1.25. Compared with correlation-based features, the RFE method produced more accurate predictions for most traits and models. Among the tested models, GA-PLS and RF showed the best performance across most traits and fractional orders, whereas XGBoost performed relatively poorly.

Fig. 8 presents the optimal accuracy of the estimation models for the six CWMs, along with the corresponding preferred strategy. These results are obtained under two band selection strategies based on fractional-order derivative spectral features. The results indicate that the optimal estimation accuracy was consistently achieved under the band selection strategy based on the RFE method. For CWM_LA, the best performance is achieved by the RF model based on the 1.00-order differential spectra, with R^2 of 0.56, an IOA of 0.84, and an RMSE of 2.75. For CWM_LT, the XGBoost model using 0.50-order differential spectra yields the highest accuracy (R^2 = 0.58, RMSE = 0.12, and IOA = 0.84). RF shows good performance for CWM_PDMC, with RF performing best at 0.50 order (R^2 = 0.60, RMSE = 0.03, and IOA = 0.84). Under 0.75-order differentiation, GA-PLS achieves the highest accuracy for CWM_TC (R^2 = 0.60, RMSE = 8.59, and IOA = 0.86), whereas, under 1.25 order, GA-PLS also performs best for CWM_TN (R^2 = 0.64, RMSE = 1.59, and IOA = 0.89). For CWM_TP, XGBoost performs the best, which, at 0.75 order, attains the best accuracy, with R^2 = 0.58, IOA = 0.85, and the lowest RMSE of 0.45.

D. Interpretability and Prediction Residuals of the Optimal Estimation Models

1) Variable Contribution Analysis of the Optimal Estimation Models: Based on the results of all the above analyses, CWM_LT, CWM_TC, and CWM_TP achieved higher prediction accuracy when using vegetation indices, with the optimal models being RF, GA-PLS, and RF, respectively. In contrast, CWM_LA, CWM_PDMC, and CWM_TN were better predicted using differential spectral features, with RF at order 1.00, RF at order 0.75, and GA-PLS at order 1.25 identified as the respective optimal models. Fig. 9 presents

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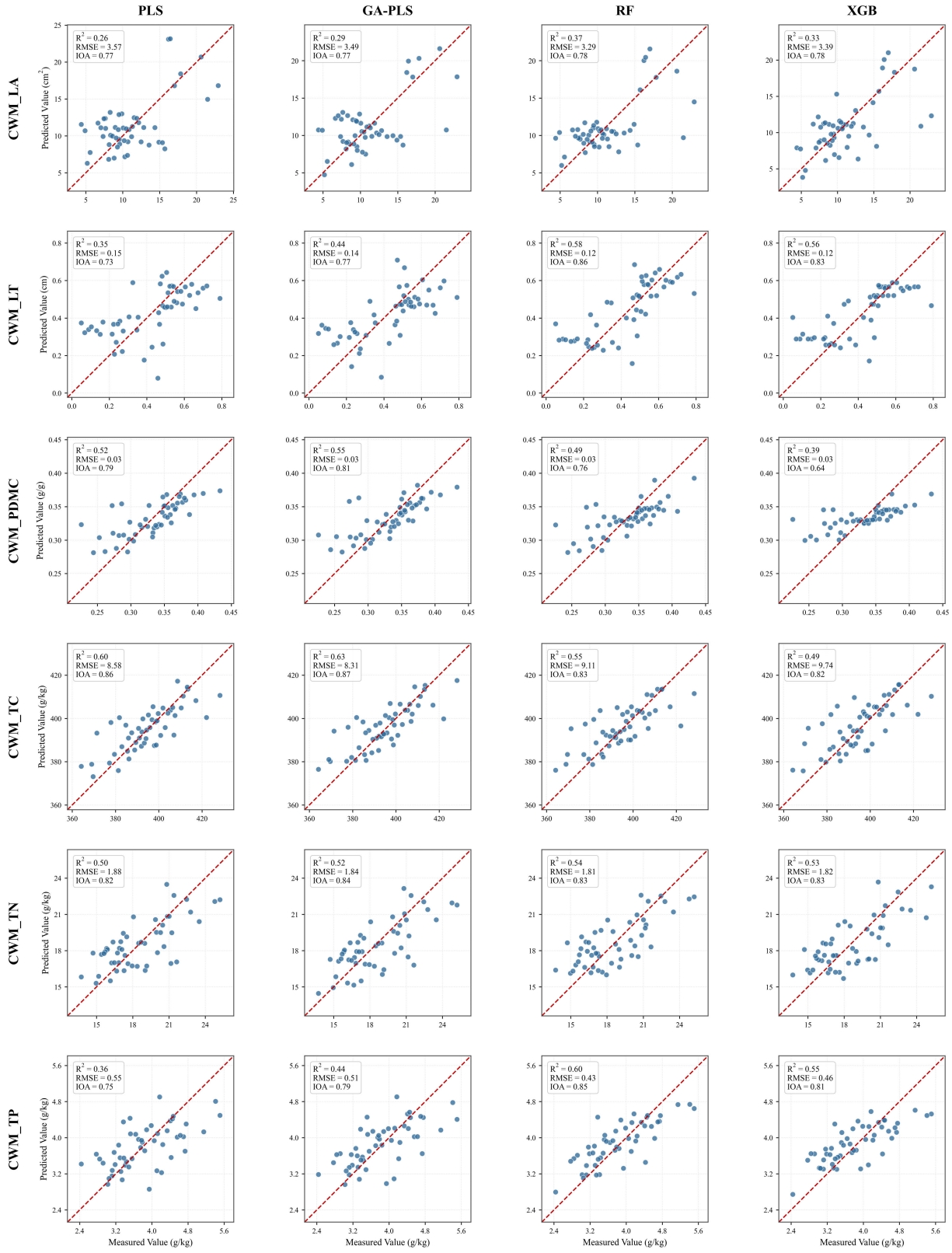


Fig. 6. Evaluation of model prediction accuracy based on vegetation indices.

the top ten variables that contribute most to the estimation results, according to the SHAP importance analysis.

For CWM_LA, reflectance in the red-edge region contributed most to the predictions, with reflectance at 718, 722, and 742 nm under the 1.00-order derivative transformation ranking as the top three contributors. For CWM_LT, NDVI and RVI exhibited the largest contributions, far exceeding those of the other vegetation indices.

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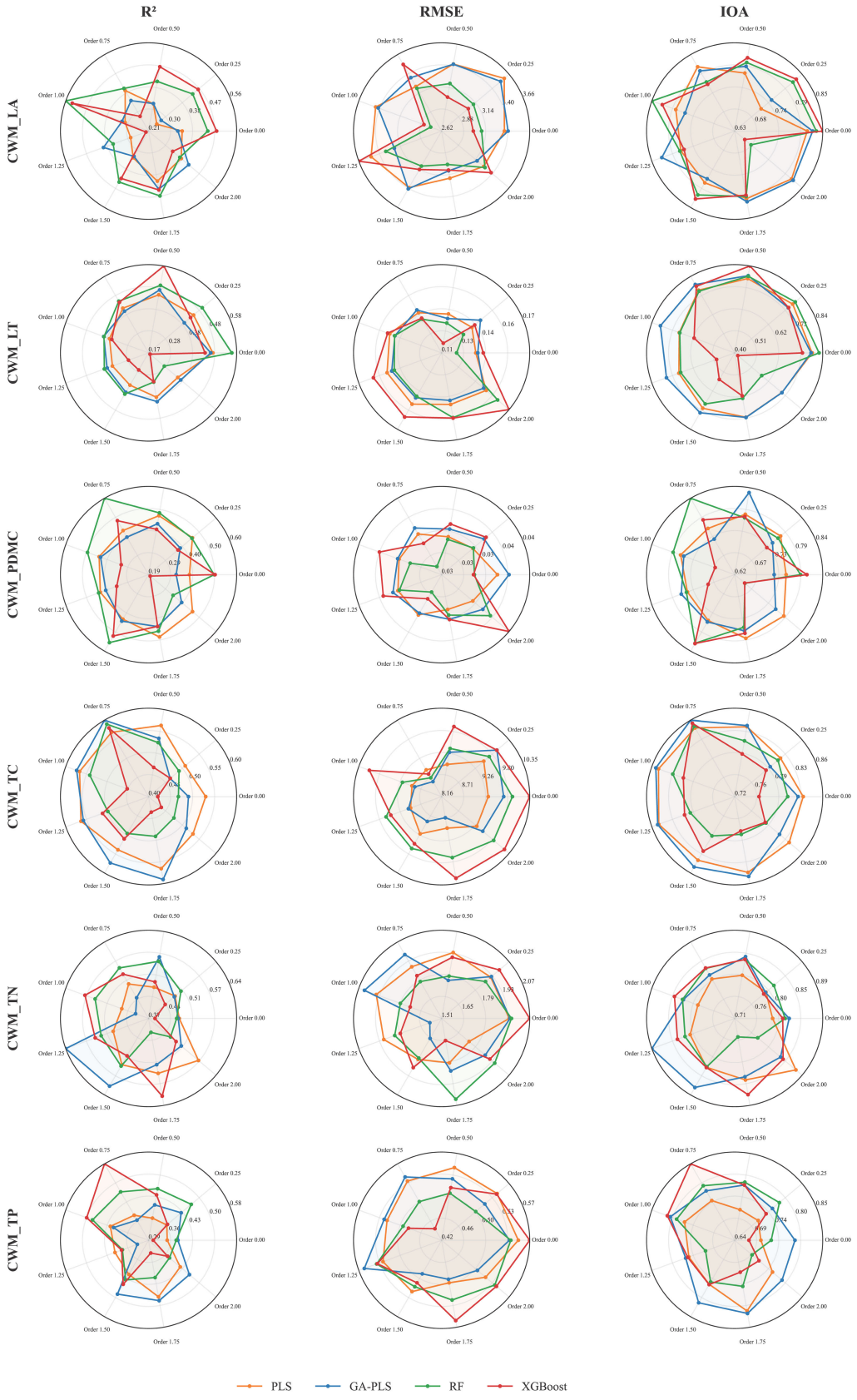


Fig. 7. Evaluation of model prediction accuracy for CWMs based on the RFE.

For CWM_PDMC, reflectance at 718 nm under the 0.75-order derivative transformation made the greatest contribution, and the highly contributing bands were mostly located in the red-edge region. For CWM_TC,

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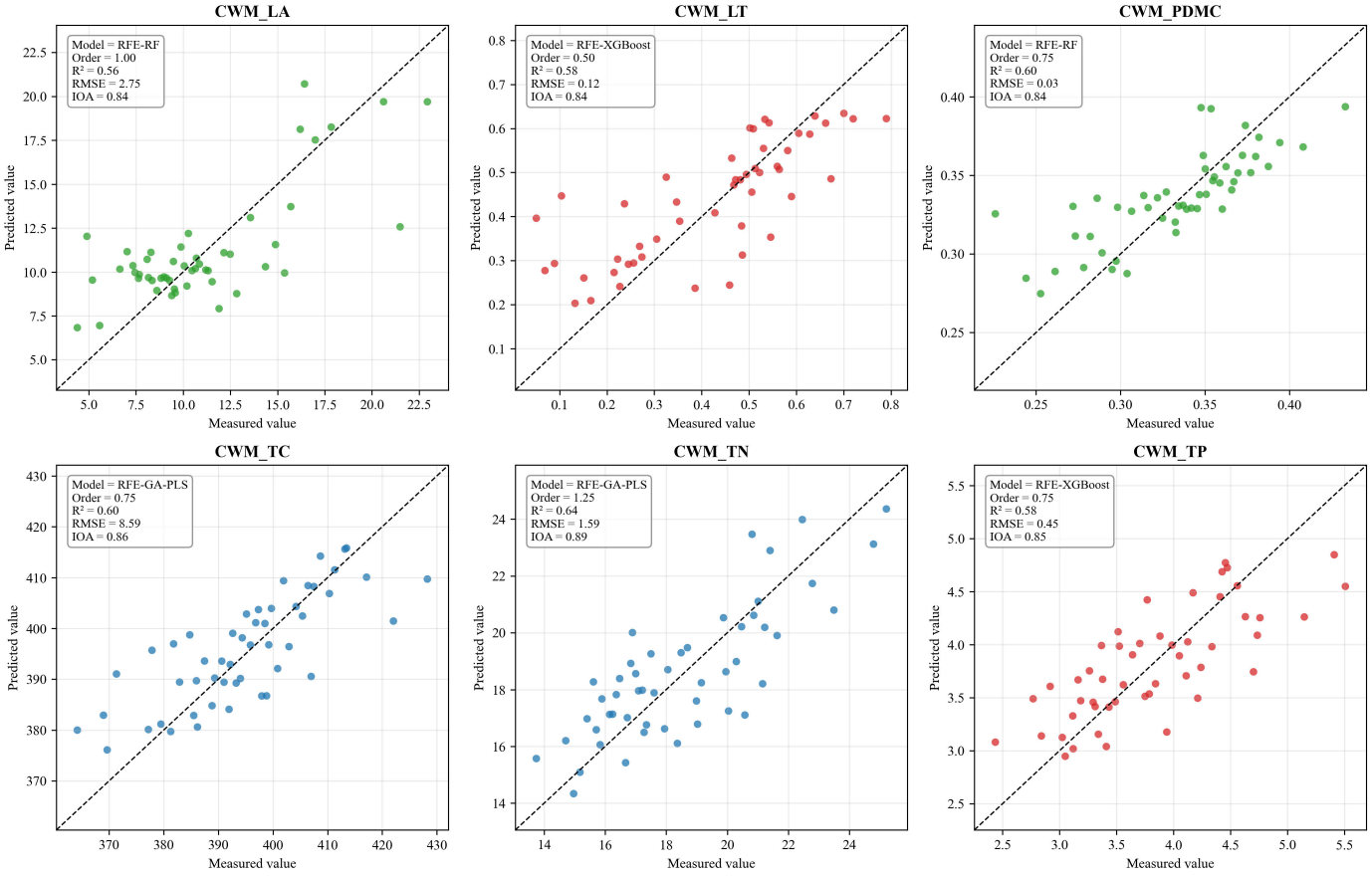


Fig. 8. Optimal estimation results for CWMs based on fractional-order differential transformations.

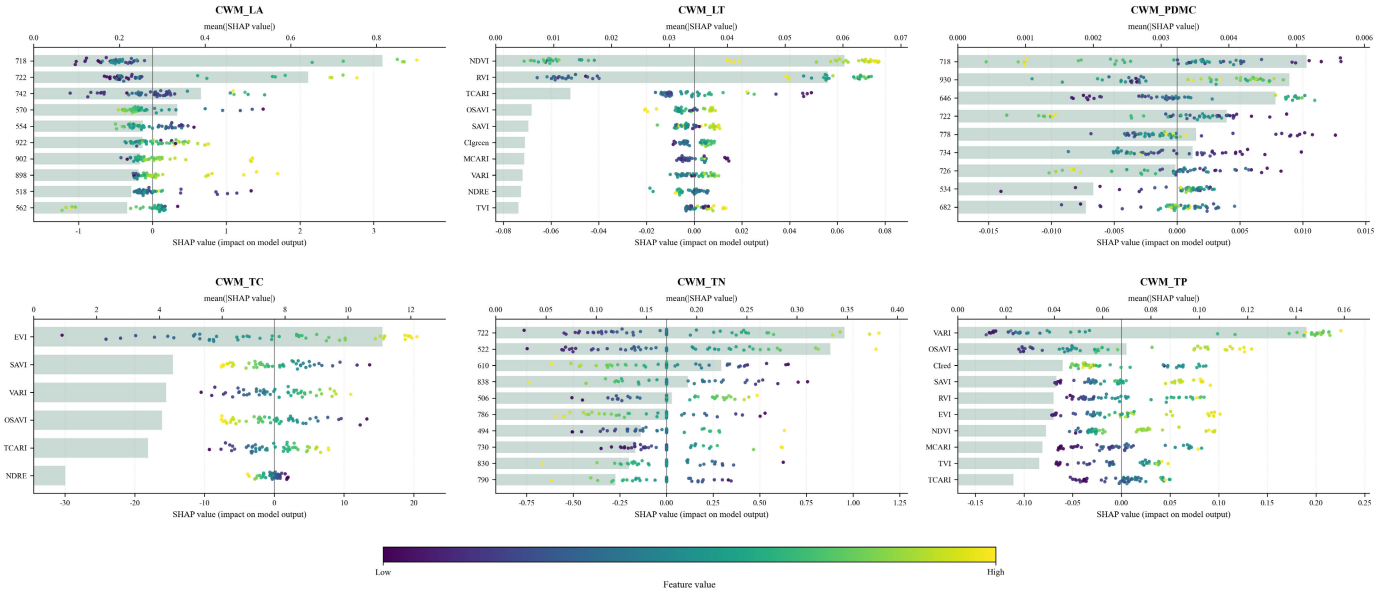


Fig. 9. Interpretation analysis of optimal estimation models using SHAP values.

EVI showed relatively high importance, while NDRE contributed the least. For CWM_TN, reflectance at 722 and 522 nm after the 1.25-order derivative treatment contributed relatively highly. For CWM_TP, VARI had the largest contribution to the predictions, whereas the importance levels of the remaining vegetation indices were relatively similar.

2) Analysis of Factors Influencing Prediction Residuals Based on the Optimal Models: To investigate potential factors influencing model errors, the prediction residuals of each CWM under its optimal prediction model were analyzed in relation to the number of dominant species within each quadrat. The results are shown in Fig. 10. Overall, no clear

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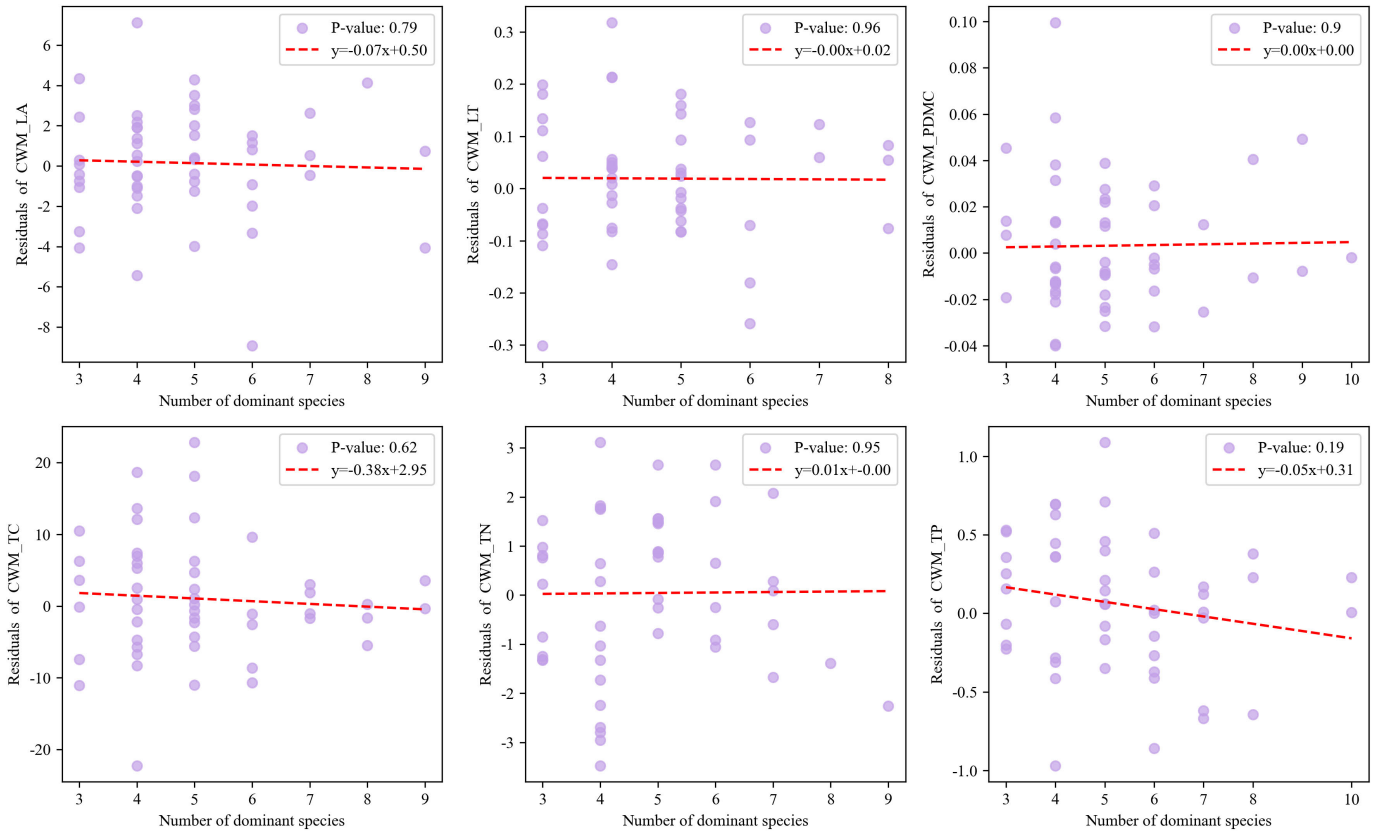


Fig. 10. Relationship between CWMs' prediction residuals and the number of dominant species.

trend or pattern in residuals was observed along the gradient of dominant species richness for any trait. Linear regression analysis was further used to quantify these relationships and to test their statistical significance. The results indicate that the relationships between prediction residuals and dominant species richness were not statistically significant for any of the traits ($P > 0.05$). This suggests that model performance is generally consistent across quadrats with different numbers of dominant species, and increases or decreases in dominant species richness do not markedly affect prediction accuracy. In other words, the optimal models exhibit good applicability and robustness across communities with varying dominant species richness.

E. Estimation of CWMs Based on Optimal Models

1) Spatial Distribution of Morphological Traits: Based on the validation accuracy of the CWM_LA, CWM_LT, and CWM_PDMC models, the best-performing model–feature combinations for each trait were used to estimate plot-scale morphological CWMs (see Fig. 11). Statistical analysis of the plot estimation results indicates that CWM_LA shows the pattern Plot3 > Plot2 > Plot1. High-value zones are mainly concentrated along the peripheries of Plot3, while Plot2 exhibits the lowest spatial heterogeneity with almost uniform low-value distribution across the entire area, and Plot1 has the lowest overall level. For CWM_LT, higher values in both Plot2 and Plot3 are concentrated in their central areas, whereas high values in Plot1 are predominantly distributed along its lower and right peripheries. For CWM_PDMC, Plot1 is generally above the overall mean with a relatively homogeneous distribution, while Plot2 and Plot3 have comparable values.

2) Spatial Distribution of Biochemical Traits: Based on the validation accuracy of the CWM_TC, CWM_TN, and CWM_TP models, the best-performing model–feature combinations for each trait were used to estimate biochemical CWMs at the plot scale (see Fig. 12). For CWM_TC, Plot2 shows the highest overall values with high concentrations in its central and upper regions, and Plot3 has high values concentrated in the middle and lower parts, with low values in the upper part. CWM_TN follows the pattern Plot2 > Plot3 > Plot1. Plot1 exhibits the lowest overall level, with high values only sporadically distributed along the edges and large contiguous low-value areas in the central region. Plot2 exhibits the highest overall level, with high values predominantly distributed along the edges. For CWM_TP, high values are concentrated in the central and upper edge regions of Plot2, whereas Plot1 is dominated by relatively low-value patches, and Plot3 exhibits an edge-high and central-low distribution pattern.

IV. Discussion

Hyperspectral remote sensing has significant potential for estimating vegetation functional traits due to its capability for rapid, large-area monitoring [34]. However, the very high spectral dimensionality also introduces challenges such as large data volumes and substantial redundancy, which may

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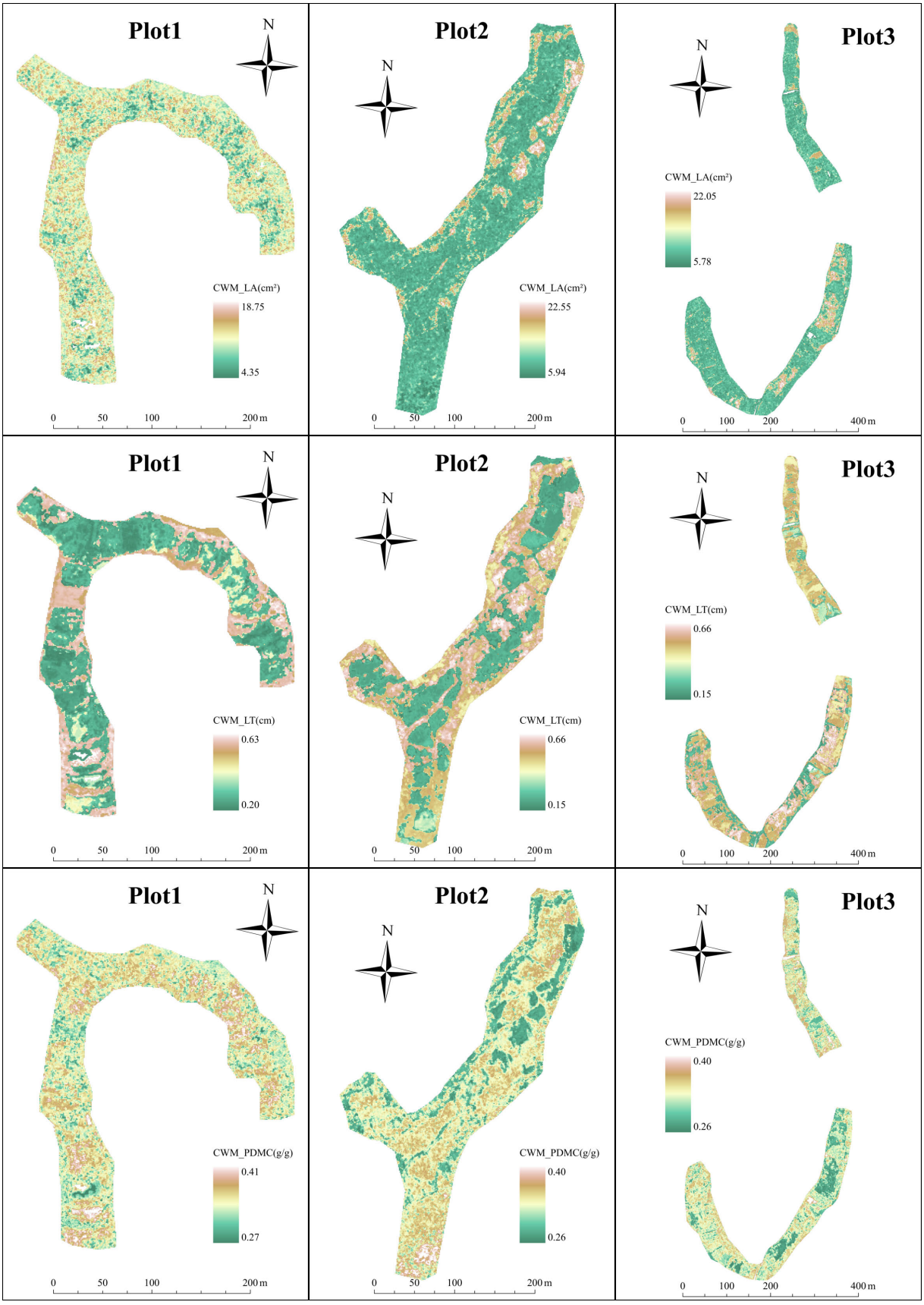


Fig. 11. Distribution of morphological traits.

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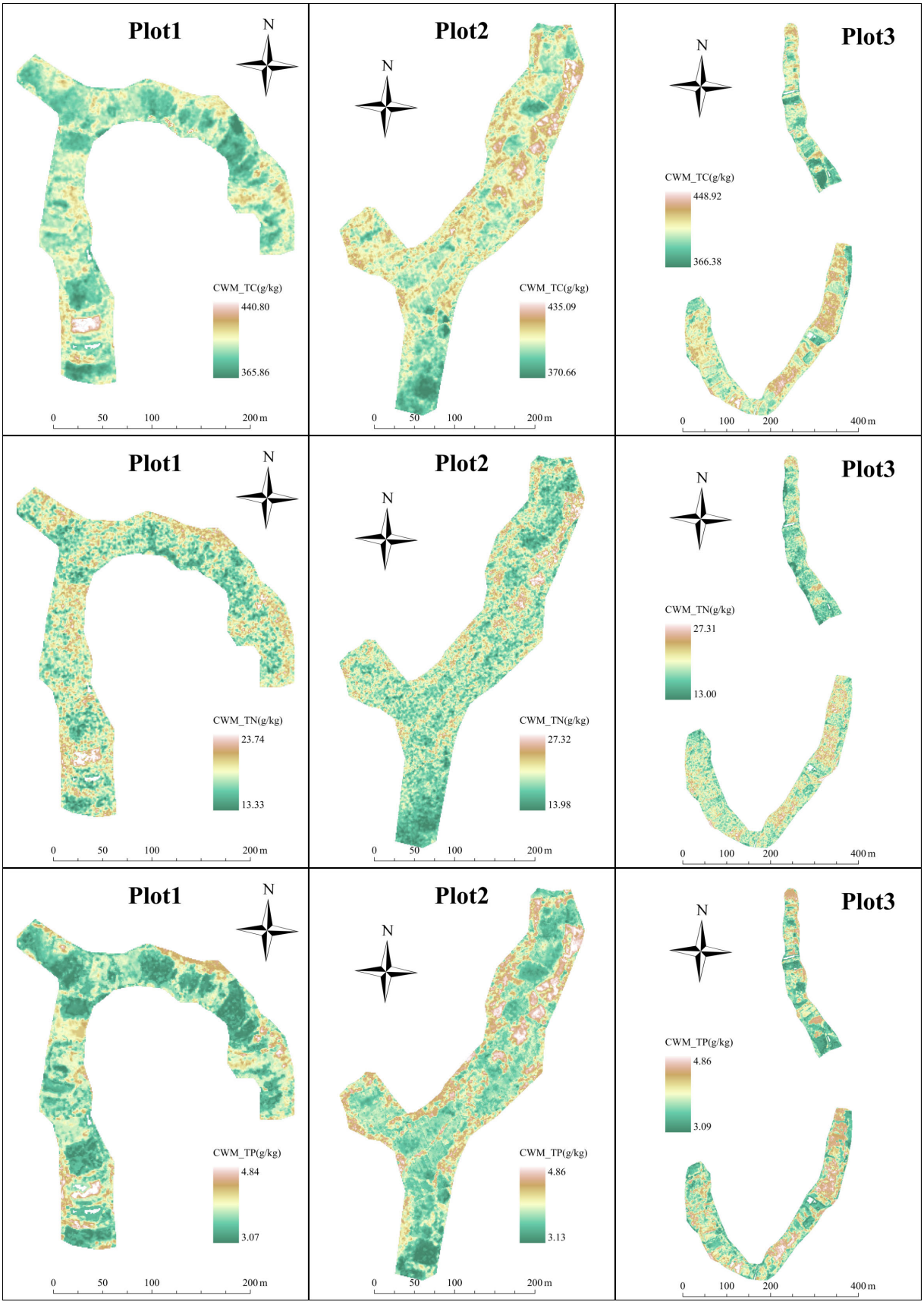


Fig. 12. Distribution of biochemical traits.

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adversely affect both estimation accuracy and computational efficiency. Consequently, extracting informative spectral features to improve model performance has become a key issue that needs to be addressed. In this study, we compared two feature strategies, namely, vegetation indices and fractional-order differential spectral features, using UAV-based hyperspectral imagery to estimate six CWMs. Overall, under appropriate feature–model combinations, all six traits achieved moderate to good prediction accuracy.

A. Advantages and Limitations of Fractional-Order Differentiation for Wetland CWM Estimation

Our results showed that fractional orders of 0.50–1.25 generally improved CWM estimation accuracy compared with the original spectra. It demonstrates that low-order FOD transformations can enhance estimation accuracy. This pattern is consistent with previous studies on grassland chlorophyll estimation [29] and leaf photosynthetic capacity parameters [35]. In addition, other studies have also verified that low-order FOD is effective in extracting ecologically relevant biophysical and biochemical parameters from hyperspectral data [36], [37], whereas the optimal fractional orders they obtained were lower than those determined in this study, which may be related to differences in research objects and observation scales.

CWM_TC, CWM_TN, and CWM_TP were more sensitive to relatively low orders, probably because their spectral signals were mainly expressed through weak absorption and red edge variations in the visible and red edge regions. Low-to-intermediate order FOD can enhance these subtle biochemical signals [37], [38], [39]. By contrast, CWM_LA and CWM_LT were more closely related to canopy structure, leaf internal structure, and multiple scattering in the near infrared region. These structural signals require differential processing at a magnitude close to the first-order derivative to achieve effective spectral enhancement and distinction [40]. CWM_PDMC may have shown an intermediate response because dry matter content is related not only to biochemical accumulation but also to leaf water status and structural properties [41].

The decline in model performance at orders above 1.25 is consistent with previous observations that excessive differentiation may amplify noise without providing additional informative features [28]. This effect is particularly important in wetland environments, where exposed water, saturated soil, canopy shadow, and mixed species canopies introduce strong background variability [39]. When the differentiation order becomes too high, local spectral fluctuations caused by these factors unrelated to plant traits may be enhanced more strongly than the CWM-related signal. Similar order-dependent responses have been reported in recent hyperspectral studies, where suitable FOD orders improved spectral details, whereas excessively high orders reduced model stability or prediction accuracy [42], [43], [44].

The results of this study indicate that vegetation indices remain effective estimators for plant physiological and biochemical parameters, which is consistent with previous findings [45], [46], [47]. However, traditional vegetation indices based on fixed-band combinations have inherent limitations and cannot fully exploit the fine spectral signals associated with target trait specificity in hyperspectral data. In contrast, optimized narrowband combinations and vegetation indices constructed based on FOD exhibit greater advantages in accurately extracting sensitive spectral features and reducing interference from complex backgrounds [13], [37], [48]. These approaches can provide more flexible and robust predictors for the high-precision estimation of CWMs, representing an important direction for future hyperspectral inversion research.

B. Model Selection and Feature Engineering for Mixed Wetland Communities

Current research linking CWMs with remote sensing data is still in its infancy, and relevant studies remain relatively scarce [49]. Therefore, more in-depth work is urgently needed to further explore their relationships. The present results provide evidence that UAV hyperspectral data can capture community-level trait variation under suitable feature selection and modeling strategies.

In this study, the prediction results for all traits showed moderate predictive ability although these accuracies were slightly lower than some previously reported results for remotely sensed plant functional traits in grassland ecosystems [50], [51]. This difference may be partly attributed to the complex spectral background of marsh wetlands. For CWM_LT, the relatively lower accuracy may be related to the scale mismatch between leaf-level thickness measurements and UAV canopy reflectance. In wetlands, canopy spectra integrate leaf traits, canopy density, species composition, shadows, water background, and saturated soil exposure. These mixed signals may weaken the direct spectral expression of leaf thickness. Among all traits, the highest R^2 values for CWM_TN and CWM_TP both exceeded 0.60, which is comparable to previous studies on remotely sensed vegetation nitrogen and physiological trait estimation [52], [53]. Recent research on wetland vegetation nitrogen and phosphorus estimation also showed that spectral matching and optimized retrieval strategies can improve CNC and CPC estimation across wetland vegetation species [39]. This indicates that nutrient-related CWMs may have more stable spectral expressions than some morphological traits under marsh wetland conditions.

The SHAP results further explained why different traits responded to different fractional orders and models. Morphological traits (CWM_LA and CWM_LT) were primarily driven by near-infrared bands from 718 to 922 nm and structural indices, such as NDVI and RVI, reflecting the sensitivity of near-infrared reflectance to canopy density, leaf internal structure, and multiple scattering processes [3], [4]. In contrast, biochemical and nutrient-related traits (CWM_TC, CWM_TN, and CWM_TP) were mainly governed by visible and red-edge bands from 610 to 722 nm and chlorophyll-related indices, including EVI, SAVI, and CIred, which are associated with pigment absorption and red-edge variations related to vegetation nitrogen and phosphorus status [52]. Traditional indices, such as NDVI and RVI, remained important because they summarize basic vegetation greenness and canopy

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density [46], although they cannot fully represent the narrowband information required for trait-specific CWM estimation.

In addition, effective feature selection, dimensionality reduction, and appropriate model selection are also critical for hyperspectral CWM estimation [54]. The GA further improved PLS by selecting informative wavelengths and reducing the influence of redundant adjacent bands, which is consistent with previous evidence that wavelength selection can improve hyperspectral PLS estimation of vegetation physiological parameters [35], [55]. This finding is also supported by recent wetland and mangrove remote sensing studies showing that multiscale feature selection and optimized machine learning can improve model performance in complex vegetation communities [56]. For the two machine learning models, XGBoost and RF, their relative performance varied among different CWMs, but, overall, RF showed slightly higher accuracy than XGBoost. The high accuracy and robustness of RF have been demonstrated in numerous studies on the quantitative estimation of wetland plant physiological and biochemical parameters [19], [57]. It is also worth noting that hyperparameter configuration is a key factor affecting the performance of both XGBoost and RF. Inappropriate hyperparameter settings may lead to overfitting or insufficient learning of critical data features. Therefore, when estimating vegetation physiological and biochemical parameters from hyperspectral data, it is essential to carefully evaluate model performance and optimize hyperparameters to improve estimation accuracy.

C. Implications for Wetland Monitoring and Future Directions

This study demonstrates the potential of UAV-based hyperspectral remote sensing for rapid and nondestructive estimation of CWMs in marsh wetlands. The achieved accuracies with R^2 values ranging from 0.56 to 0.64 are comparable to those reported [53], [58]. Considering the complex spectral background of wetlands, including mixed species composition, exposed water, saturated soil, and heterogeneous canopy structure, these results indicate that UAV hyperspectral data can provide useful information for monitoring wetland plant functional traits.

However, several limitations should be considered. Mixed pixels containing vegetation, water, and saturated soil may alter canopy reflectance and weaken the relationship between spectra and CWMs [5]. First, the water background was not fully corrected in this study. In this study, we only adopted a threshold of NDVI = 0.4 based on previous literature to reduce the influence of environmental background interference, without further exploring or optimizing the optimal threshold value. In addition, future studies could use water-adjusted vegetation indices or spectral unmixing methods to reduce the influence of water background [59]. Second, this study was conducted at a single period, while wetland plant traits and community composition may change with phenology, hydrological conditions, and seasonal growth stages [50].

Although hyperspectral techniques have been proven effective for estimating wetland vegetation functional traits [10], [15], the estimation of morphological CWMs remains challenging because CWM_LA and CWM_LT are easily affected by canopy structure, species mixture, and background exposure. Previous studies have shown that multisource remote sensing can improve the estimation of vegetation traits [60], [61], [62]. Since hyperspectral data provide limited structural information, combining them with LiDAR-derived canopy height and vertical structure metrics may help improve the estimation accuracy of structural traits [63]. Similar multisource remote sensing strategies have also been applied to wetland vegetation mapping and biomass estimation [64], [65]. In addition, recent studies have shown that transferring hyperspectral trait estimation methods to satellite imagery is feasible, providing new insights for regional-scale vegetation trait monitoring [66]. The current plot-scale models still require further validation before regional application. Future studies should include more wetland types, larger sample sizes, and independent validation sites and further evaluate model transferability across different wetland types and spatial scales.

V. Conclusion

This study used UAV-based hyperspectral imagery to estimate six CWMs in marsh wetland plant communities. Vegetation indices and fractional-order differential spectral features were extracted, and four models, including PLS, GA-PLS, RF, and XGBoost, were compared. The main conclusions are given as follows.

1) UAV hyperspectral data showed good potential for estimating CWMs in marsh wetlands. Under suitable feature-model combinations, all six CWMs achieved moderate predictive ability, with optimal R^2 values ranging from 0.56 to 0.64.
2) Fractional-order differentiation improved spectral feature extraction and CWM estimation. Low-to-intermediate orders, especially 0.50–1.25, were more effective than the original spectra and high-order transformations, while excessively high orders reduced model stability and prediction accuracy.
3) Different CWMs showed trait-specific responses to spectral features and models. CWM_LA, CWM_PDMC, and CWM_TN were better estimated using fractional-order differential spectra, whereas CWM_LT, CWM_TC, and CWM_TP were better estimated using vegetation indices.

References

[1] C. Violle et al., "Let the concept of trait be functional!" *Oikos*, vol. 116, no. 5, pp. 882–892, May 2007, doi: 10.1111/j.0030-1299.2007.15559.x.
[2] H. Moor, H. Rydin, K. Hylander, M. B. Nilsson, R. Lindborg, and J. Norberg, "Towards a trait-based ecology of wetland vegetation," *J. Ecol.*, vol. 105, no. 6, pp. 1623–1635, Nov. 2017, doi: 10.1111/1365-2745.12734.
[3] S. L. Ustin et al., "Retrieval of foliar information about plant pigment systems from high resolution spectroscopy," *Remote Sens. Environ.*, vol. 113, pp. S67–S77, Sep. 2009, doi: 10.1016/j.rse.2008.10.019.
[4] G. P. Asner and R. E. Martin, "Spectranomics: Emerging science and conservation opportunities at the interface of biodiversity and remote sensing," *Global Ecology Conservation*, vol. 8, pp. 212–219, Oct. 2016, doi: 10.1016/j.gecco.2016.09.010.

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[5] K. Klančnik, I. Zelnik, P. Gnezda, and A. Gaberščik, "Do reflectance spectra of different plant stands in wetland indicate species properties?" in *The Role of Natural and Constructed Wetlands in Nutrient Cycling and Retention on the Landscape*. Cham, Switzerland: Springer, 2015, pp. 73–86.

[6] K. Klančnik and A. Gaberščik, "Leaf spectral signatures differ in plant species colonizing habitats along a hydrological gradient," *J. Plant Ecol.*, vol. 9, no. 4, pp. 442–450, Aug. 2016, doi: 10.1093/jpe/rtv068.

[7] J. Pacheco-Labrador et al., "Challenging the link between functional and spectral diversity with radiative transfer modeling and data," *Remote Sens. Environ.*, vol. 280, Oct. 2022, Art. no. 113170, doi: 10.1016/j.rse.2022.113170.

[8] W. W. Sun et al., "Research progress and prospects of hyperspectral remote sensing for global wetland from 2010 to 2022," *Natl. Remote Sens. Bull.*, vol. 27, no. 6, pp. 1281–1299, 2023, doi: 10.11834/jrs.20232620.

[9] Y. L. Yao, X. Wang, X. P. Tan, and Y. Q. Shan, "Models building between leaf spectral reflectance and traits for typical marshland wetland plants based on PLSR: A case of Sanjiang national nature reserve," *Scientia Geographica Sinica*, vol. 42, no. 9, pp. 1638–1645, 2022, doi: 10.13249/j.cnki.sgs.2022.09.013.

[10] A. J. Rebelo, B. Somers, K. J. Esler, and P. Meire, "Can wetland plant functional groups be spectrally discriminated?" *Remote Sens. Environ.*, vol. 210, pp. 25–34, Jun. 2018, doi: 10.1016/j.rse.2018.02.031.

[11] Z. Dou et al., "Hyperspectral inversion of suaeda salsa biomass under different types of human activity in liaohe estuary wetland in northeastern China," *Mar. Freshwater Res.*, vol. 71, no. 4, pp. 482–492, Mar. 2020, doi: 10.1071/mf19030.

[12] W. Zheng, X. Lu, Y. Li, S. Li, and Y. Zhang, "Hyperspectral identification of chlorophyll fluorescence parameters of suaeda salsa in coastal wetlands," *Remote Sens.*, vol. 13, no. 11, p. 2066, 2021, doi: 10.3390/rs13112066.

[13] X. Jiang et al., "Newly-developed three-band hyperspectral vegetation index for estimating leaf relative chlorophyll content of mangrove under different severities of pest and disease," *Ecol. Indicators*, vol. 140, Jul. 2022, Art. no. 108978, doi: 10.1016/j.ecolind.2022.108978.

[14] J. Wang, Y. Xu, and G. Wu, "The integration of species information and soil properties for hyperspectral estimation of leaf biochemical parameters in mangrove forest," *Ecol. Indicators*, vol. 115, Aug. 2020, Art. no. 106467, doi: 10.1016/j.ecolind.2020.106467.

[15] W. Li et al., "Predictions of spartina alterniflora leaf functional traits based on hyperspectral data and machine learning models," *Eur. J. Remote Sens.*, vol. 57, no. 1, Dec. 2024, Art. no. 2294951, doi: 10.1080/22797254.2023.2294951.

[16] W. Li et al., "Suitability of hyperspectral data for monitoring nitrogen and phosphorus content in constructed wetlands," *Remote Sens. Lett.*, vol. 11, no. 5, pp. 495–504, May 2020, doi: 10.1080/2150704x.2020.1734247.

[17] C. Guo and X. Guo, "Estimating leaf chlorophyll and nitrogen content of wetland emergent plants using hyperspectral data in the visible domain," *Spectrosc. Lett.*, vol. 49, no. 3, pp. 180–187, Mar. 2016, doi: 10.1080/00387010.2015.1118126.

[18] L. Cui et al., "Hyperspectral inversion of phragmites communis carbon, nitrogen, and phosphorus stoichiometry using three models," *Remote Sens.*, vol. 12, no. 12, p. 1998, Jun. 2020, doi: 10.3390/rs12121998.

[19] X. Tang et al., "Hyperspectral prediction of mangrove leaf stoichiometries in different restoration areas based on machine learning models," *J. Appl. Remote Sens.*, vol. 16, no. 3, Aug. 2022, Art. no. 034525, doi: 10.1117/1.jrs.16.034525.

[20] L. Wang and Y. Wei, "Revised normalized difference nitrogen index (NDNI) for estimating canopy nitrogen concentration in wetlands," *Optik*, vol. 127, no. 19, pp. 7676–7688, Oct. 2016, doi: 10.1016/j.ijleo.2016.05.115.

[21] H. Z. Li et al., "Estimating aboveground biomass of wetland plant communities from hyperspectral data based on fractional-order derivatives and machine learning," *Remote Sens.*, vol. 16, no. 16, p. 3011, Aug. 2024, doi: 10.3390/rs16163011.

[22] J. Sandino, G. Pegg, F. Gonzalez, and G. Smith, "Aerial mapping of forests affected by pathogens using UAVs, hyperspectral sensors, and artificial intelligence," *Sensors*, vol. 18, no. 4, p. 944, Mar. 2018, doi: 10.3390/s18040944.

[23] H. Ren, Y. Zhao, W. Xiao, and Z. Hu, "A review of UAV monitoring in mining areas: Current status and future perspectives," *Int. J. Coal Sci. Technol.*, vol. 6, no. 3, pp. 320–333, Sep. 2019, doi: 10.1007/s40789-019-00264-5.

[24] W. Zhuo et al., "Aboveground biomass retrieval of wetland vegetation at the species level using UAV hyperspectral imagery and machine learning," *Ecol. Indicators*, vol. 166, Sep. 2024, Art. no. 112365, doi: 10.1016/j.ecolind.2024.112365.

[25] H. Gholizadeh, J. A. Gamon, A. I. Zygielbaum, R. Wang, A. K. Schweiger, and J. Cavender-Bares, "Remote sensing of biodiversity: Soil correction and data dimension reduction methods improve assessment of α-diversity (species richness) in Prairie ecosystems," *Remote Sens. Environ.*, vol. 206, pp. 240–253, Mar. 2018, doi: 10.1016/j.rse.2017.12.014.

[26] C. Xu et al., "Assessing the impact of soil on species diversity estimation based on UAV imaging spectroscopy in a natural Alpine steppe," *Remote Sens.*, vol. 14, no. 3, p. 671, Jan. 2022, doi: 10.3390/rs14030671.

[27] J. Wang, L. Cui, W. Gao, T. Shi, Y. Chen, and Y. Gao, "Prediction of low heavy metal concentrations in agricultural soils using visible and near-infrared reflectance spectroscopy," *Geoderma*, vol. 216, pp. 1–9, Mar. 2014, doi: 10.1016/j.geoderma.2013.10.024.

[28] X. Ge et al., "Estimating agricultural soil moisture content through UAV-based hyperspectral images in the arid region," *Remote Sens.*, vol. 13, no. 8, p. 1562, Apr. 2021, doi: 10.3390/rs13081562.

[29] A. Zhang, S. Yin, J. Wang, N. He, S. Chai, and H. Pang, "Grassland chlorophyll content estimation from drone hyperspectral images combined with fractional-order derivative," *Remote Sens.*, vol. 15, no. 23, p. 5623, Dec. 2023, doi: 10.3390/rs15235623.

[30] J. Wang et al., "Desert soil clay content estimation using reflectance spectroscopy preprocessed by fractional derivative," *PLoS ONE*, vol. 12, no. 9, Sep. 2017, Art. no. e0184836, doi: 10.1371/journal.pone.0184836.

[31] A. Ghosh and P. K. Joshi, "A comparison of selected classification algorithms for mapping bamboo patches in lower gangetic plains using very high resolution WorldView 2 imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 26, pp. 298–311, Feb. 2014, doi: 10.1016/j.jag.2013.08.011.

[32] P. Sharma, "Evaluating tree explanation methods for anomaly reasoning: A case study of SHAP TreeExplainer and TreeInterpreter," in *Proc. 39th Int. Conf. Conceptual Model. (ER)*, 2020, pp. 35–45, doi: 10.1007/978-3-030-65847-2_4.

[33] W. Huang, W. Li, J. Xu, X. Ma, C. Li, and C. Liu, "Hyperspectral monitoring driven by machine learning methods for grassland aboveground biomass," *Remote Sens.*, vol. 14, no. 9, p. 2086, Apr. 2022, doi: 10.3390/rs14092086.

[34] Y. Zhang, M. Migliavacca, J. Penuelas, and W. Ju, "Advances in hyperspectral remote sensing of vegetation traits and functions," *Remote Sens. Environ.*, vol. 252, Jan. 2021, Art. no. 112121, doi: 10.1016/j.rse.2020.112121.

[35] G. Song, Q. Wang, and J. Jin, "Fractional-order derivative spectral transformations improved partial least squares regression estimation of photosynthetic capacity from hyperspectral reflectance," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510110, doi: 10.1109/TGRS.2023.3270892.

[36] Y. Hong et al., "Combination of fractional order derivative and memory-based learning algorithm to improve the estimation accuracy of soil organic matter by visible and near-infrared spectroscopy," *CATENA*, vol. 174, pp. 104–116, Mar. 2019, doi: 10.1016/j.catena.2018.10.051.

[37] Y. Xu et al., "Improving photosynthetic pigment estimation in leaves using low-order fractional derivative spectral indices: Development and validation of synthetic and experimental data," *Int. J. Remote Sens.*, vol. 46, no. 18, pp. 6783–6802, Sep. 2025, doi: 10.1080/01431161.2025.2546153.

[38] S. Bhadra et al., "Quantifying leaf chlorophyll concentration of sorghum from hyperspectral data using derivative calculus and machine learning," *Remote Sens.*, vol. 12, no. 13, p. 2082, Jun. 2020, doi: 10.3390/rs12132082.

[39] Z. Lao et al., "A novel constrained spectral matching approach for extending UAV multispectral reflectance measurements and estimating nitrogen and phosphorus contents in wetland vegetation species," *Plant Phenomics*, vol. 7, no. 2, Jun. 2025, Art. no. 100059, doi: 10.1016/j.plaphe.2025.100059.

[40] Y. Liu et al., "Estimation of potato above-ground biomass using UAV-based hyperspectral images and machine-learning regression," *Remote Sens.*, vol. 14, no. 21, p. 5449, Nov. 2022, doi: 10.3390/rs14215449.

[41] I. J. Wright et al., "The worldwide leaf economics spectrum," *Nature*, vol. 428, no. 6985, pp. 821–827, Apr. 2004, doi: 10.1038/nature02403.

[42] B. Yang et al., "Estimation of chlorophyll content of cinnamomum camphora leaves based on hyperspectral and fractional order differentiation," *Int. J. Remote Sens.*, vol. 45, no. 15, pp. 5113–5129, Aug. 2024, doi: 10.1080/01431161.2024.2372064.

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[43] J. Zhang et al., "Monitoring soil salinization in arid cotton fields using unmanned aerial vehicle hyperspectral imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 140, Jun. 2025, Art. no. 104584, doi: 10.1016/j.jag.2025.104584.

[44] L. Xiao, Y. Ma, X. Gao, B. Song, and L. Wu, "Multiscale feature enhancement and lightweight ensemble modeling for hyperspectral chlorophyll inversion in greenhouse tomato," *Smart Agricult. Technol.*, vol. 13, Mar. 2026, Art. no. 101885, doi: 10.1016/j.atech.2026.101885.

[45] S. Cui and K. Zhou, "A comparison of the predictive potential of various vegetation indices for leaf chlorophyll content," *Earth Sci. Informat.*, vol. 10, no. 2, pp. 169–181, Jun. 2017, doi: 10.1007/s12145-016-0281-3.

[46] L. Caturegli et al., "Normalized difference vegetation index versus dark green colour index to estimate nitrogen status on bermudagrass hybrid and tall fescue," *Int. J. Remote Sens.*, vol. 41, no. 2, pp. 455–470, Jan. 2020, doi: 10.1080/01431161.2019.1641762.

[47] J. Zhang et al., "Leaf area index estimation model for UAV image hyperspectral data based on wavelength variable selection and machine learning methods," *Plant Methods*, vol. 17, no. 1, p. 49, Dec. 2021, doi: 10.1186/s13007-021-00750-5.

[48] X. Tong, L. Duan, T. Liu, Z. Yang, Y. Wang, and V. P. Singh, "Estimation of grassland aboveground biomass combining optimal derivative and raw reflectance vegetation indices at peak productive growth stage," *Geocarto Int.*, vol. 38, no. 1, Dec. 2023, Art. no. 2186497, doi: 10.1080/10106049.2023.2186497.

[49] O. J. A. Martínez et al., "Scaling up functional traits for ecosystem services with remote sensing: Concepts and methods," *Ecol. Evol.*, vol. 6, no. 13, pp. 4359–4371, Jul. 2016, doi: 10.1002/ece3.2201.

[50] Z. Wang et al., "Mapping foliar functional traits and their uncertainties across three years in a grassland experiment," *Remote Sens. Environ.*, vol. 221, pp. 405–416, Feb. 2019, doi: 10.1016/j.rse.2018.11.016.

[51] Y.-W. Zhang et al., "Estimating community-level plant functional traits in a species-rich Alpine meadow using UAV image spectroscopy," *Remote Sens.*, vol. 14, no. 14, p. 3399, Jul. 2022, doi: 10.3390/rs14143399.

[52] A. Ramoelo, A. K. Skidmore, M. A. Cho, M. Schlerf, R. Mathieu, and I. M. A. Heitkönig, "Regional estimation of savanna grass nitrogen using the red-edge band of the spaceborne RapidEye sensor," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 19, pp. 151–162, Oct. 2012, doi: 10.1016/j.jag.2012.05.009.

[53] Y. Zhao et al., "Hyperspectral retrieval of leaf physiological traits and their links to ecosystem productivity in grassland monocultures," *Ecol. Indicators*, vol. 122, Mar. 2021, Art. no. 107267, doi: 10.1016/j.ecolind.2020.107267.

[54] J. P. Rivera-Caicedo, J. Verrelst, J. Muñoz-Marí, G. Camps-Valls, and J. Moreno, "Hyperspectral dimensionality reduction for biophysical variable statistical retrieval," *ISPRS J. Photogramm. Remote Sens.*, vol. 132, pp. 88–101, Oct. 2017, doi: 10.1016/j.isprsjprs.2017.08.012.

[55] T. Mehmood, S. Sæbø, and K. H. Liland, "Comparison of variable selection methods in partial least squares regression," *J. Chemometrics*, vol. 34, no. 6, Jun. 2020, Art. no. e3226, doi: 10.1002/cem.3226.

[56] B. Fu et al., "Quantifying scattering characteristics of mangrove species from optuna-based optimal machine learning classification using multi-scale feature selection and SAR image time series," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 122, Aug. 2023, Art. no. 103446, doi: 10.1016/j.jag.2023.103446.

[57] E. Adam, O. Mutanga, E. M. Abdel-Rahman, and R. Ismail, "Estimating standing biomass in papyrus (Cyperus papyrus L.) swamp: Exploratory of in situ hyperspectral indices and random forest regression," *Int. J. Remote Sens.*, vol. 35, no. 2, pp. 693–714, Jan. 2014, doi: 10.1080/01431161.2013.870676.

[58] C. Li, H. Wulf, B. Schmid, J.-S. He, and M. E. Schaepman, "Estimating plant traits of Alpine grasslands on the Qinghai-Tibetan Plateau using remote sensing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 7, pp. 2263–2275, Jul. 2018, doi: 10.1109/JSTARS.2018.2824901.

[59] P. Villa, A. Mousivand, and M. Bresciani, "Aquatic vegetation indices assessment through radiative transfer modeling and linear mixture simulation," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 30, pp. 113–127, Aug. 2014, doi: 10.1016/j.jag.2014.01.017.

[60] S. Cooper, A. Okujeni, D. Pflugmacher, S. van der Linden, and P. Hostert, "Combining simulated hyperspectral EnMAP and Landsat time series for forest aboveground biomass mapping," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 98, Jun. 2021, Art. no. 102307, doi: 10.1016/j.jag.2021.102307.

[61] C. Chen, Y. Ma, G. Ren, and J. Wang, "Aboveground biomass of salt-marsh vegetation in coastal wetlands: Sample expansion of in situ hyperspectral and Sentinel-2 data using a generative adversarial network," *Remote Sens. Environ.*, vol. 270, Mar. 2022, Art. no. 112885, doi: 10.1016/j.rse.2021.112885.

[62] J. L. Pancorbo et al., "Airborne hyperspectral and sentinel imagery to quantify winter wheat traits through ensemble modeling approaches," *Precis. Agricult.*, vol. 24, no. 4, pp. 1288–1311, Aug. 2023, doi: 10.1007/s11119-023-09990-y.

[63] S. Luo et al., "Retrieving aboveground biomass of wetland phragmites Australis (common reed) using a combination of airborne discrete-return LiDAR and hyperspectral data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 58, pp. 107–117, Jun. 2017, doi: 10.1016/j.jag.2017.01.016.

[64] Y. Du, J. Wang, Z. Liu, H. Yu, Z. Li, and H. Cheng, "Evaluation on spaceborne multispectral images, airborne hyperspectral, and LiDAR data for extracting spatial distribution and estimating aboveground biomass of wetland vegetation suaeda salsa," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 1, pp. 200–209, Jan. 2019, doi: 10.1109/JSTARS.2018.2886046.

[65] B. Fu et al., "Quantifying vegetation species functional traits along hydrologic gradients in Karst wetland based on 3D mapping with UAV hyperspectral point cloud," *Remote Sens. Environ.*, vol. 307, Jun. 2024, Art. no. 114160, doi: 10.1016/j.rse.2024.114160.

[66] B. Fu et al., "Cross-scenario transfer learning for estimating mangrove nitrogen and phosphorus content from field hyperspectral data to SDGSAT-1 and sentinel-2 images," *Remote Sens. Environ.*, vol. 329, Nov. 2025, Art. no. 114923, doi: 10.1016/j.rse.2025.114923.

[67] Y. Liu, S. Liu, J. Li, X. Guo, S. Wang, and J. Lu, "Estimating biomass of winter oilseed rape using vegetation indices and texture metrics derived from UAV multispectral images," *Comput. Electron. Agricult.*, vol. 166, Nov. 2019, Art. no. 105026, doi: 10.1016/j.compag.2019.105026.

[68] Y. Liu et al., "Improving potato above ground biomass estimation combining hyperspectral data and harmonic decomposition techniques," *Comput. Electron. Agricult.*, vol. 218, Mar. 2024, Art. no. 108699, doi: 10.1016/j.compag.2024.108699.

[69] A. Kross, H. McNairn, D. Lapen, M. Sunohara, and C. Champagne, "Assessment of RapidEye vegetation indices for estimation of leaf area index and biomass in corn and soybean crops," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 34, pp. 235–248, Feb. 2015, doi: 10.1016/j.jag.2014.08.002.

[70] Y. Zhang et al., "Estimating the maize biomass by crop height and narrowband vegetation indices derived from UAV-based hyperspectral images," *Ecol. Indicators*, vol. 129, Oct. 2021, Art. no. 107985, doi: 10.1016/j.ecolind.2021.107985.



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